# Measuring the Photoluminescence of Semiconducting Nanoparticles

Alessandro Ciavatta

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**Abstract**

Photoluminescence measurement are performed on three $CdS_xSe_{(1-x)}$/ZnS quantum dots samples, with different concentration $x$ of Sulphur and Selenium. Theese materials have a tunable energy gap, that varies with the dimension of the nanocrystals and with the concentration. The emission spectrum of the samples is characterized by a peak centered in the first excited state to ground state transition energy, which is in the visible range for all the three samples taken in analysis. Some of the estimated concentration have non-physical value, this is probably due to the large uncertainty of the measures and to the ignorance on the geometric structure of the nanocrystals.

## Introduction

In a Photoluminescence (PL) experiment, a semiconductor sample is irradiated by photons with energy greater than the energy gap of the sample. The photon is able to excite an electron into the conduction band, leaving an hole in the valence band. The electron (hole) will move rapidly into the bottom (top) of the conduction (valence) band via phonon interaction; therefore the recombination happens when the energy is near the energy gap of the material, with a spread of $k_BT$. The transition are considered vertical: the photons involved have a trascurable momentum with respect to the momentum of the pair ($\lambda_{ph} \gg a$), where $a$ is the lattice spacing, of the order of the electron's de Broglie wavelength. If the dimension of the crystal is reduced, the electrons experience quantum confinement: electron momentum become quantized in the direction of confinement and discontinuities into the density of states appear. The most confined structrure is the zero-dimensional quantum dot (QD), in which the density of states is a sum Dirac's delta funcions, as in atomic electronic structure. The confinement well potential, which has the form of the particle-in-a-box potential, is obtained surrounding a semiconductor nanocrystal with another semiconductor with wider energy gap. If the gap energy of the outer semiconductor is much greater, the infinite potential approximation is valid, therefore the allowed energies are the energies of a particle-in-a-box of length equal to the diameter of the QD and mass equal to the effective mass in the band taken into account. For a cubic confining potential with length $L$, the energy spectrum is

$$E_g + \frac{h^2 n^2}{8L^2}\left(\frac{1}{m_e^*} + \frac{1}{m_h^*}\right),$$

where the excited states of both the conduction and valence band are considered. In a PL experiment, the radiative emission comes primarely from the lowest energy excited state.

## Materials and Methods

The samples analysed are QD with different composition and approximately the same diameter (6 nm)

and geometry. The confinement barrier is obtained with ZnS, the dots are made of $CdS_xSe_{1-x}$, with $0 \leq x \leq 1$. The QD samples are excited with 395 nm UV diode and a 375 nm UV laser, emitting in the visible range. The output photons are collected into a spectrometer via an optic fiber. The UV peak due to the source is always present: in order to acquire a spectrum where the peak of the sample is visible, we can change: the direction of the source; the integration time, i.e. the time of acquisition; the luminosity of the source. Another sample taken in analysis is a perovskite crystal, namely $MAPbBr_3$; recently studied for its optoelectronic properties.

## Results and Discussion

The emission spectrum of three samples (red, green, blue) containing different concentration of S and Se are measured, via diode and via laser. The emission spectrum of the samples does not consist of a narrow peak centered at the emission energy, some broadening effects are always present. The natural broadening, due to uncertainty principle, results in a Lorentzian distribution; while thermal broadening has a gaussian profile. Assuming that the length of the QD is normally distributed, there is also contribution that is not gaussian, nor Lorentzian. For simplicity, the fit is gaussian, but a deviation from the gaussian shape is visible in Fig. (1). In the blue sample seems to be present also a low and broad peak of yellow color, probably due to contamination. Assuming that the nanoparticles are of the same shape and size, and assuming that the PL peaks are due to the transition from the first excited state to the ground state, we can estimate the concentration $x$ of the three samples. $m_e^*$, $m_h^*$ and $E_g$ depend on $x$:

$$
\begin{aligned}
m_e^* &= x m_e^*(CdS) + (1-x) m_e^*(CdSe) \\
m_h^* &= x m_h^*(CdS) + (1-x) m_h^*(CdSe) \\
E_g &= x E_g(CdS) + (1-x) E_g(CdSe) + bx(1-x),
\end{aligned}
$$

where $b = 0.24$ eV is known as bowing parameter. Tab. (1) explicits all the other parameters needed (at 300 K). The estimated concentration $x$ depends strongly on the size and geometry of the QD. For instance, if the QD is cubic with length $L$, $n^2 = n_x^2 + n_y^2 + n_z^2 = 3$ for the first excited state; if the QD is spherical with radius $L$, $n^2 = 1$. Tab. (2) summarizes the results.

| | $E_g$ (eV) | $m_e^*/m_e$ | $m_h^*/m_e$ |
|---|---|---|---|
| CdSe | 1.74 | 0.13 | 0.45 |
| CdS | 2.45 | 0.21 | 0.8 |

*Table 1: Parameters needed to estimate the concentration $x$ [1] [2].*

| $x$ / geometry | red | green | blue |
|---|---|---|---|
| cubic (L=6nm) | -0.27 | 0.26 | 0.99 |
| spherical (r=3nm) | -0.47 | 0.12 | 0.84 |
| spherical (r=6nm) | 0.00 | 0.51 | 1.33 |

*Table 2: Estimated values of $x$. $L$ is the length of the cube, $r$ is the radius of the sphere.*

PL measurement via UV laser is performed on a perovskite $MAPbBr_3$ crystal. The corresponding energy peak is found at $(2.14 \pm 0.03)$. The complete spectrum is plotted in Fig. (2).

## Conclusions

Because of our ignorance in the geometric structure of the QD, it is difficult to estimate the correct concentration of S and Se in the three samples. Moreover, the standard deviation of the peaks results in an high uncertainty in the concentration measures. The PL spectrum of the perovskite crystal is noisier because the PL emission is less intense, but a peak in the yellow visible range is isolated.

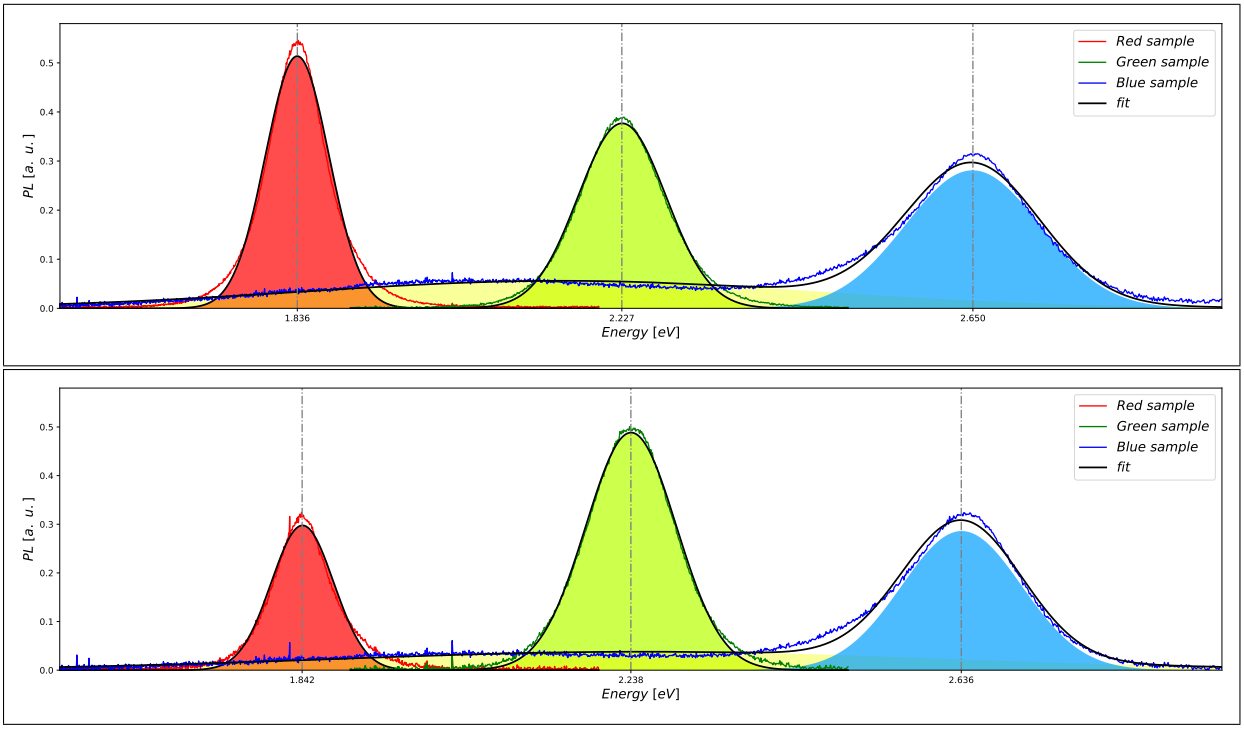



*Figure 1: PL spectrum of three QD samples with different compositions. The source is the UV led in the upper spectrum and laser in the lower spectrum. The three spectrum are taken from separate data acquisition. The RGB color correspond exactly to the wavelength of the peak.*

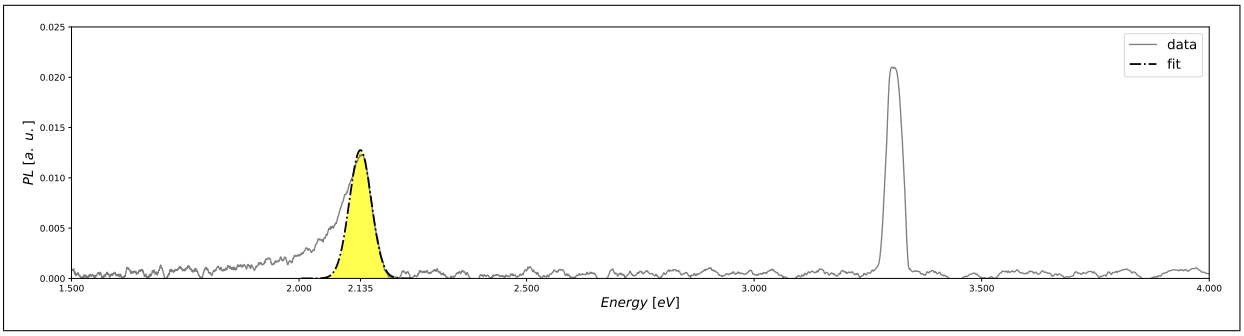



*Figure 2: PL spectrum of perovskite $MAPbBr_3$ crystal. The peak on the right is due to the U laser. The RGB color correspond exactly to the wavelength of the peak.*

# References


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[2] R. Dalven. "Calculation of effective masses in cubic CdS and CdSe". In: *physica status solidi (b)* 48.1 (1971), K23–K26. DOI: https://doi.org/10.1002/pssb.2220480150. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/pssb.2220480150. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/pssb.2220480150.